



Fast Interpretable Greedy-Tree Sums

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Modern machine learning has achieved impressive prediction performance, but often sacrifices interpretability, a critical consideration in high-stakes domains such as medicine. In such settings, practitioners often use highly interpretable decision tree models, but these suffer from inductive bias against additive structure. To overcome this bias, we propose Fast Interpretable Greedy-Tree Sums (FIGS), which generalizes the Classification and Regression Trees (CART) algorithm to simultaneously grow a flexible number of trees in summation. By combining logical rules with addition, FIGS adapts to additive structure while remaining highly interpretable. Experiments on real-world datasets show FIGS achieves state-of-the-art prediction performance. To demonstrate the usefulness of FIGS in high-stakes domains, we adapt FIGS to learn clinical decision instruments (CDIs), which are tools for guiding decision-making. Specifically, we introduce a variant of FIGS known as Group Probability-Weighted Tree Sums (G-FIGS) that accounts for heterogeneity in medical data. G-FIGS derives CDIs that reflect domain knowledge and enjoy improved specificity (by up to 20% over CART) without sacrificing sensitivity or interpretability. Theoretically, we prove that FIGS learns components of additive models, a property we refer to as disentanglement. Further, we show (under oracle conditions) that tree-sum models leverage disentanglement to generalize more efficiently than single tree models when fitted to additive regression functions. Finally, to avoid overfitting with an unconstrained number of splits, we develop Bagging-FIGS, an ensemble version of FIGS that borrows the variance reduction techniques of random forests. Bagging-FIGS performs competitively with random forests and XGBoost on real-world datasets.

interpretability | XGBoost | clinical decision instrument | CART | random forest

Modern machine learning methods such as random forests (1), gradient boosting (2, 3), and deep learning (4) display impressive predictive performance, but are complex and opaque, leading many to call them “black-box” models. Model interpretability is critical in many applications (5, 6), particularly in high-stakes settings such as clinical decision instrument (CDI) modeling. Interpretability allows models to be audited for general validation, errors, or biases, and therefore also more amenable to improvement by domain experts. Further, interpretability facilitates counterfactual reasoning, which is the foundation of scientific insight, and it instills trust/distrust in a model when warranted. As an added benefit, interpretable models tend to be faster and more computationally efficient than black-box models.*

Decision trees are a prime example of interpretable models (2, 7–10). They can be easily visualized, memorized, and emulated by hand, even by nonexperts, and thus fit naturally into high-stakes use-cases, such as decision-making in medicine (e.g., the emergency department[†]), law, and public policy. While decision trees have the potential to adapt to complex data, they are often outperformed by black-box models in terms of prediction performance. However, there is evidence that this performance gap is not intrinsic to interpretable models (7, 10–12). In this paper, we identify an inductive bias of decision trees that causes its prediction performance to suffer in some instances and design a tree-based algorithm that overcomes this bias while preserving interpretability.

Our starting point is the observation that decision trees can be statistically inefficient at fitting regression functions with additive components (13). To illustrate this, consider the following toy example: $y = \mathbb{1}_{X_1 > 0} + \mathbb{1}_{X_2 > 0} \cdot \mathbb{1}_{X_3 > 0}$.[‡] The two components of this

Significance

Machine learning holds the potential to solve many real-world problems, but interpretability is a necessary prerequisite for practitioners in high-stakes domains such as medicine and law. Decision trees are mostly interpretable, but often do not have the best accuracy. Here, we propose a type of tree-based model, called Fast Interpretable Greedy-tree Sums (FIGS), that aims to produce accurate and interpretable models. FIGS extends Classification and Regression Trees (CART), an existing algorithm that builds a single decision tree, to instead build a sum of trees with a chosen cap on the total number of rules. This extension allows FIGS to capture additive structure in data, achieving better accuracy than CART while remaining interpretable across a variety of real-world datasets.

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*FIGS is integrated into the imodels package github.com/csinva/imodels (7) with an sklearn-compatible API. Experiments for reproducing the results here can be found at github.com/Yu-Group/imodels-experiments.

[†]For example, in the popular tool [mdcalc](https://www.mdcalc.com/), over 90% of available CDIs take the form of a decision tree.

[‡]This toy model is an instance of a Local Spiky and Sparse model (14), which is grounded in real biological mechanisms whereby an outcome is driven by interactions of inputs (e.g. biomolecules) which display thresholding behavior.

function can be individually implemented by trees with 1 split and 2 splits respectively. However, fitting their sum with a single tree requires at least 5 splits, as we are forced to make *duplicate subtrees*: A copy of the second tree has to be grown out of every leaf node of the first tree (Fig. 1). Indeed, given independent tree functions $f_1, \dots, f_k$, in order for a single tree f to implement their sum, we would generally need

$$\#splits(f) \geq \prod_{k=1}^K (\#splits(f_k) + 1) - 1.$$

In such a setting, a single decision tree is an inferior model compared to the tree-sum because it requires more splits. As argued in refs. 13 and 15, the number of splits is strongly related to the variance of the fitted model. Second, we argue in this paper that the number of splits is a good measure of the model's level of interpretability. Furthermore, the single decision tree obscures true interactions in the generative model—we are unable to tell, based on the tree structure, which features belong to the same additive component of the generative model.

Ideally, we would like to be able to fit a separate tree to each additive component of the generative mechanism and present the *tree-sum* as our fitted model. Searching for an appropriate tree-sum model, however, is a difficult task. Indeed, while existing ensemble methods such as random forests (1) and gradient boosted trees (2, 3) comprise tree-sums, they either fit each tree independently (random forests) or sequentially (gradient boosting), and are hence unable to disentangle additive components.

To address this gap in the literature, we propose Fast Interpretable Greedy-Tree Sums (FIGS), a natural algorithm that is able to *grow a flexible number of trees simultaneously*. FIGS is based on a simple yet effective modification to Classification and Regression Trees (CART) (8), allowing it to adapt to additive structure (if present) by starting new trees, while still maintaining the ability of CART to adapt to higher-order interaction terms. By capping the total number of splits allowed, FIGS produces a model that is also easily visualized, memorized, and emulated by hand.

We perform extensive experiments across a wide array of real-world datasets to compare the predictive performance of FIGS to a number of popular decision rule models (16). Specifically, we take the number of rules (splits in the case of trees) as a common measure of interpretability for this model class, and construct decision rule models at a prescribed level of interpretability. Our results show FIGS often achieves the best predictive performance across various levels of interpretability (i.e., number of splits).

In order to provide insight into the predictive performance of FIGS, we conduct additional experiments ([SI Appendix, section S2](#)) in which we apply FIGS and other decision rule models to a number of commonly simulated datasets.[§] Our experiments show that, in comparison to other methods, FIGS' increased flexibility and adaptivity leads to lower bias and prediction error, at the cost of slightly higher variance and a larger proportion of splits on noisy features.

Next, we apply FIGS to a key domain problem involving decision rule models which is the construction of CDIs. CDIs are models for predicting patient risk and are widely used by

healthcare professionals to make rapid decisions in a clinical setting such as the emergency room. To apply FIGS to the medical domain, it is crucial that FIGS is able to adapt to heterogeneous data from diverse groups of patients. Tailoring models to various groups is often necessary since various groups of patients may differ dramatically and require distinct features for high predictive performance on the same outcome. While a naive solution is to fit a unique model to each group of patients, this is sample-inefficient and sacrifices valuable information that can be shared across groups. To mitigate this issue, we introduce a variant of FIGS, called Group Probability-Weighted Tree Sums (G-FIGS), which accounts for this heterogeneity while using all the samples available. Specifically, G-FIGS first fits a classifier (e.g., logistic regression) to predict group membership probabilities for each sample. Then, it uses these estimates as instance weights in FIGS to output a model for each group.

We use both FIGS and G-FIGS to construct CDIs for three pediatric emergency care datasets and found that both have up to 20% higher specificity than CART while maintaining the same level of sensitivity. Further, G-FIGS improves specificity over FIGS by over 3% for fixed levels of sensitivities. The features used in the fitted models agree with medical domain knowledge. Moreover, G-FIGS learns a tree-sum model where each tree in the model represents a distinct clinical domain, providing a clear and organized framework for clinicians to use when assessing and treating patients. Next, we investigate the stability of G-FIGS to data perturbations. Stability to “reasonable” data perturbations is a key tenet of the Predictability, Computability, and Stability (PCS) framework (6, 20, 21), and a necessary prerequisite for the application of ML techniques in high-stakes domains. We show that G-FIGS learns a similar tree-sum model (i.e., a similar set of features) across data perturbations (e.g., introducing noise by randomly permuting labels). This work expands on initial results on G-FIGS contained in an earlier preprint (22).

Next, to provide insight into the success of FIGS, we investigate tree-sum models and FIGS theoretically. We prove generalization upper bounds for tree-sum models when given oracle access to the optimal tree structures. Whenever additive structure is present, this upper bound has a faster rate in the sample size n compared to the generalization lower bound for any decision tree proved in ref. 13. Further, in the large-sample limit FIGS disentangles the additive components of the generative model, with each tree in the sum fitting a separate additive component.

Last, similarly to CART, FIGS overfits when allowed too many splits. Hence, we propose an ensemble version, called Bagging-FIGS, that borrows the bootstrap and feature subsetting strategies of random forests. Bagging-FIGS is competitive with random forests, XGBoost, and generalized additive models (GAMs) across a wide range of real-world datasets. However, ensemble methods like Bagging-FIGS and random forests lack the interpretability of single or sum of decision tree models.

In what follows, Section 1 introduces FIGS and Section 2 covers related work. Section 3 contains our experimental results on real-world datasets showing that FIGS predicts well with very few splits. Section 4 covers three CDI case studies using FIGS and G-FIGS. Section 5 investigates the theoretical performance of tree-sum models and FIGS. Finally, Section 6 introduces Bagging-FIGS and compares its prediction performance to other algorithms on real-world datasets.

[§]We generated i.i.d. Gaussian covariate data and responses using a number of commonly studied analytic regression functions. i.i.d. covariates are commonly used to study feature selection and bias-variance tradeoffs of tree-based methods (17–19).

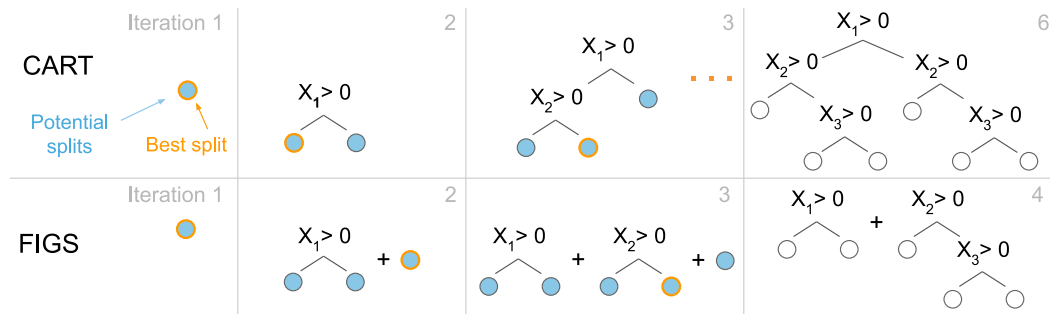


Fig. 1. FIGS algorithm overview for learning the toy function $y = \mathbb{1}_{X_1 > 0} + \mathbb{1}_{X_2 > 0} \cdot \mathbb{1}_{X_3 > 0}$. FIGS greedily adds one node at a time, considering splits not just in an individual tree but within a collection of trees in a sum. This can lead to much more compact models, as it reduces repeated splits (e.g., in the final CART model shown in the *Top-Right*).

1. FIGS: Algorithm Description and Run-Time

Suppose we are given training data $\mathcal{D}_n = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$. When growing a tree, CART chooses for each node t the split s that maximizes the impurity decrease in the responses $\mathbf{y}$. For a given node t , the impurity decrease has the expression

$$\hat{\Delta}(s, t, \mathbf{y}) := \sum_{\mathbf{x}_i \in t} (y_i - \bar{y}_t)^2 - \sum_{\mathbf{x}_i \in t_L} (y_i - \bar{y}_{t_L})^2 - \sum_{\mathbf{x}_i \in t_R} (y_i - \bar{y}_{t_R})^2,$$

where t_L and t_R denote the left and right child nodes of t respectively, and $\bar{y}_t, \bar{y}_{t_L}, \bar{y}_{t_R}$ denote the mean responses in each of the nodes. We call such a split s a *potential split*, and note that for each iteration of the algorithm, CART chooses the potential split with the largest impurity decrease.[†]

FIGS extends CART to greedily grow a tree-sum (Algorithm 1). That is, at each iteration of FIGS, the algorithm chooses either to make a split on one of the current K trees $\hat{f}_1, \dots, \hat{f}_K$ in the sum, or to add a new stump to the sum. To make this decision, it still applies the CART splitting criterion detailed above to identify a potential split in each leaf of each tree. However, to compute the impurity decrease for a given split, it substitutes the vector of *residuals* $r_i := y_i - \sum_{k=1}^K \hat{f}_k(\mathbf{x}_i)$ for the vector of responses $\mathbf{y}$.[#] FIGS makes only one split among the $K + 1$ trees: the one corresponding to the largest impurity decrease. The value of each of the new leaf nodes is then defined to be the mean residual for the samples it contains, added to the value of its parent node. If a new stump is created, the value at the root is defined to be zero. At inference time, we predict the response of an example by dropping it down each tree and summing the values of each leaf node containing it.

1.1. Selecting the Model's Stopping Threshold. Choosing a threshold on the total number of splits can be done similarly to CART: using a combination of the model's predictive performance (i.e., cross-validation) and domain knowledge on how interpretable the model needs to be. Alternatively, the threshold can be selected using an impurity decrease threshold (8) rather than a hard threshold on the number of splits. Potential data-driven choices of the threshold in Section 7.

[†]This corresponds to greedily minimizing the mean-squared-error criterion in regression and Gini impurity in binary classification. For multiclass classification, each mean-squared-error term is replaced by the Gini impurity.

[#]Since the current tree makes a constant prediction over the node being split, one may ignore its contributions to the residuals without affecting the impurity decrease.

Algorithm 1 FIGS fitting algorithm.

```

1: FIGS( $X$ : features,  $y$ : outcomes, stopping_threshold)
2:  $trees = []$ 
3: while count_total_splits( $trees$ ) < stopping_threshold:
4:    $all\_trees = \text{join}(trees, \text{build\_new\_tree}())$  # add new tree
5:    $potential\_splits = []$ 
6:   for  $tree$  in  $all\_trees$ :
7:      $y\_residuals = y - \text{predict}(all\_trees)$ 
8:     for  $leaf$  in  $tree$ :
9:        $potential\_split = \text{split}(X, y\_residuals, leaf)$ 
10:       $potential\_splits.append(potential\_split)$ 
11:    $best\_split = \text{split\_with\_min\_impurity}(potential\_splits)$ 
12:    $trees.insert(best\_split)$ 

```

1.2. Run-Time Analysis. The run-time complexity for FIGS to grow a model with m splits in total is $O(dm^2n^2)$, where d the number of features, and n the number of samples (see derivation in *SI Appendix, section S1*). In contrast, CART has a run-time of $O(dmn^2)$. Since only a small number of splits is made to ensure interpretability, the difference between run times is negligible.

1.3. Extensions. FIGS supports many natural modifications that are used in Classification and Regression Trees. For example, different impurity measures can be used; here we use Gini impurity for classification and mean-squared-error for regression. Additionally, FIGS could benefit from pruning,[‡] shrinkage (23), or by being used as part of an ensemble model (e.g., Bagging-FIGS). Other extensions are discussed in *SI Appendix, section S1*.

2. Related Work and Its Connections to FIGS

There is a long history of greedy methods for learning individual trees, e.g., C4.5 (9), CART (8), and ID3 (24). Recent work has proposed global optimization procedures rather than greedy algorithms for trees; these can improve performance given a fixed split budget but incur a high computational cost (25–27). However, due to the limitations of a single tree, these methods have an inductive bias against additive structure (13, 28); FIGS avoids these limitations by simultaneously growing an additive ensemble of trees. Besides trees, there are a variety of other interpretable methods such as rule lists

[‡]Typical pruning strategies used in CART can be used with FIGS with one modification: After pruning a leaf, residuals for all other trees must be updated before pruning the leaf. This substantially increases the computational complexity of pruning.

(29, 30), rule sets (31–33), or generalized additive models (34, 35); for an overview and Python implementations, see ref. 7.

FIGS is related to *boosting* (36–38), but differs in that it allows i) adaptable tree sizes and ii) the ability to selectively revisit previous trees after fitting new trees. These two changes allow FIGS to make better use when the number of splits are limited. A detailed discussion of the differences between FIGS and boosting in *SI Appendix, section S3*. FIGS is similarly related to *backfitting* (34, 39), but differs in that it 1) again does not assume a fixed number of component features and 2) does not require knowledge on which features are used by each component. Furthermore, FIGS does not update its component trees in a cyclic manner, but instead trees “compete for splits” at each iteration.

Similar to the work here are methods that learn an additive model of splits, where a split is defined to be an axis-aligned, rectangular region in the input space. RuleFit (40) is a popular method that learns a model by first extracting splits from multiple greedy decision trees fit to the data and then learning a linear model using those splits as features. FIGS improves upon RuleFit by greedily selecting higher-order interactions when needed, rather than simply using all splits from some prespecified tree depth. MARS (41) greedily learns an additive model of splines in a manner similar to FIGS, but loses some interpretability as a result of using splines rather than splits. FIGS is related to the base estimator proposed in the random planted forest (42), but differs in that FIGS i) restricts a model’s total number of splits rather than the number of unique features used in each tree, and ii) does not use feature subsampling, enabling it to make better use of a limited rule budget.

Also related to this work are tree ensembles, such as random forest (1), gradient-boosted trees (36), Bayesian Additive Regression Trees (BART) (43), AddTree (44), and Groves (45), all of which use ensembling as a way to boost predictive accuracy without focusing on finding an interpretable model. Loosely related are post hoc methods which aim to help understand a black-box model (2, 46, 47), particularly tree-specific feature importances such as Mean Decrease In Impurity (MDI) and Mean Decrease in Accuracy (MDA) (1, 15, 48–50). However, these can display lack of fidelity to the original model, and also suffer from other problems with feature correlations and summarizing interactions (51).

3. FIGS Results on Real-World Benchmark Datasets

This section shows that FIGS enjoys strong prediction performance on several real-world benchmark datasets compared to popular algorithms for fitting decision rule models.

3.1. Benchmark Datasets. For classification, we study four large datasets previously used to evaluate rule-based models (23, 52) along with the two largest University of California, Irvine (UCI) binary classification datasets used in Breiman’s original paper introducing random forests (1, 53) (overview in Table 1). For regression, we study all datasets used in the random forest paper with at least 200 samples along with three of the largest nonredundant datasets from the Penn Machine Learning Benchmark benchmark (57). 80% of the data is used for training/threefold cross-validation and 20% of the data is used for testing.

3.2. Baseline Methods. For both classification and regression, FIGS is compared to CART, RuleFit, and Boosted Stumps

Table 1. Real-world datasets analyzed here: Classification (Top panel), regression (Bottom panel)

	Name	Samples	Features	Majority class
Classification	Readmission	101,763	150	53.9%
	Credit (54)	30,000	33	77.9%
	Recidivism	6,172	20	51.6%
	Juvenile (55)	3,640	286	86.6%
	German credit	1,000	20	70.0%
	Diabetes (56)	768	8	65.1%
	Name	Samples	Features	Mean
Regression	Breast tumor (57)	116,640	9	62.0
	CA housing (58)	20,640	8	5.0
	Echo months (57)	17,496	9	74.6
	Satellite image (57)	6,435	36	7.0
	Abalone (59)	4,177	8	29.0
	Diabetes (60)	442	10	346.0
	Friedman1 (41)	200	10	26.5
	Friedman2 (41)	200	4	1,657.0
	Friedman3 (41)	200	4	1.6

(CART stumps learned via gradient-boosting).** Furthermore, for classification, we additionally compare against C4.5. For regression, we additionally compare against a CART model fitted using the mean-absolute-error splitting-criterion. We finally also add a black-box baseline comprising a Random Forest with 100 trees, which thus uses many more splits than all the other models.†† See full details on models and hyperparameters in *SI Appendix, section S4*.

3.3. FIGS Predicts Well with Few Splits. Fig. 2 shows the models’ test set performance as a function of the number of splits in the fitted model.‡‡ We treat the number of splits as a common metric for the level of interpretability of each model. In practice, interpretability is context-dependent. However, the goal of this experiment is to sweep across a range of interpretability levels, and show the test performance had that level been selected.

The top two rows of Fig. 2 show results for classification [measured using the area under the receiver operating characteristic (ROC) curve, i.e., AUC], and the bottom three rows show results for regression (measured using the coefficient of determination, denoted by R^2). On average, FIGS outperforms baseline models when the number of splits is very low. The performance gain from FIGS over other baselines is larger for the datasets with more samples (e.g., the *Top* row of Fig. 2), matching the intuition that FIGS performs better because of its increased flexibility. For two of the larger datasets (*Credit* and *Recidivism*), FIGS even outperforms the black-box Random Forest baseline, despite using less than 15 splits. For the smallest classification dataset (*Diabetes*), FIGS performs extremely well with very few (less than 10) splits but starts to overfit as more splits are added.

3.4. FIGS has Smaller Bias on a Range of Simulated DGPs. We also study the comparative performance of FIGS on simulated datasets across a number of regression functions. Across all regression functions, FIGS had smaller bias and mean squared

**We have omitted results on boosted trees of larger depth to avoid clutter. In our experiments, we found setting larger depth to decrease prediction performance. See *SI Appendix, section S3* for these experiments.

††We also attempt to compare to optimal tree methods, such as Generalized Optimal Sparse Decision Trees (25), but find that they are unable to fit the dataset sizes here.

‡‡For RuleFit, each term in the linear model is counted as one split.

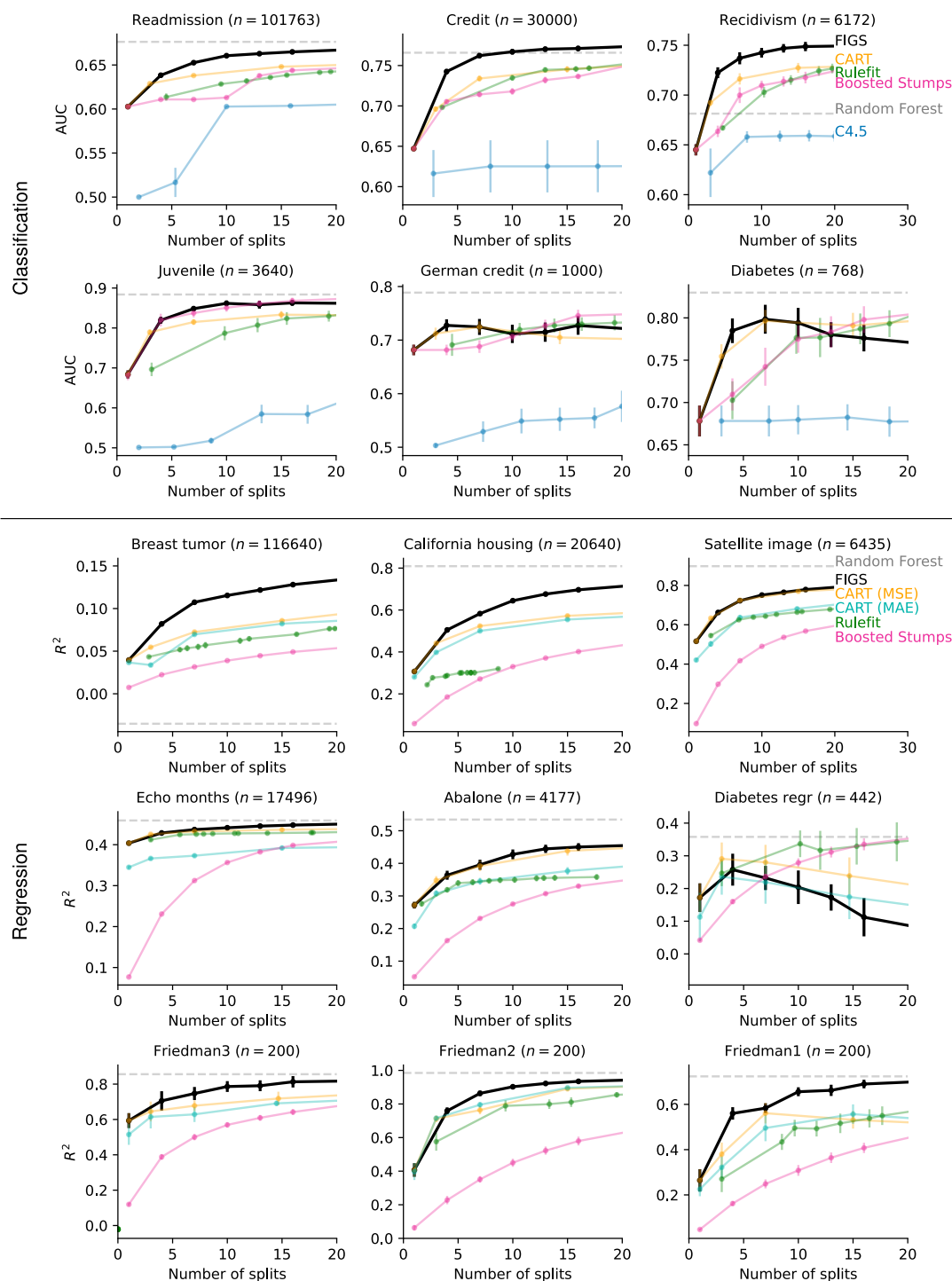


Fig. 2. FIGS performs extremely well on the test-set using very few splits, particularly when the dataset is large. *Top* two rows show results for classification datasets (measured by AUC of the ROC curve) and the *Bottom* three rows show results for regression datasets (measured by R^2). Errors bars show SE of the mean, computed over 6 random data splits.

error compared with CART, Boosted Stumps, and Gradient Boosted Trees of depth 2. On the other hand, each other method had high bias for at least one choice of regression function. The results are discussed in *SI Appendix, section S2*.

4. Learning CDIs via FIGS

An important use-case of interpretable models is the development of CDIs, which are evidence-based tools for assisting clinicians

in their decision-making process. Recent works have developed and validated CDIs using interpretable models, particularly in emergency medicine (61–64). One challenge in applying machine learning models to the medial domain is to account for heterogeneous subpopulations (e.g., age groups, sex, treatment sites). Typically, this heterogeneity is dealt by fitting a separate model on each group. However, this strategy comes at the cost of losing samples, and discarding valuable information that can be shared among various groups. To mitigate this loss of power, we

introduce a variant of FIGS, called group-probability weighted FIGS, or G-FIGS as follows.

4.1. G-FIGS: Fitting FIGS to Multiple Groups. As before, consider a supervised learning setting with features X , outcome Y . Additionally, assume we have access to group label G (e.g., treatment site or age group) for each sample. G-FIGS is a two-step algorithm: i) For a given group label g , G-FIGS first estimates group membership probabilities for each sample (e.g., by fitting a logistic regression model to predict group-membership)^{§§} That is, it estimates $\mathbb{P}(G = g | X)$. ii) For a given group g , G-FIGS then uses the group probabilities $\mathbb{P}(G = g | X)$ as sample weights when fitting FIGS to the *whole dataset*. This results in a group-specific model, but borrows information from samples in other groups. See [SI Appendix, section S5](#) for more details, and a visual representation.

4.2. Datasets and Data Cleaning. Table 2 shows the CDI datasets under consideration here. They each constitute a large-scale multisite data aggregation by the Pediatric Emergency Care Applied Research Network (PECARN), with a relevant clinical outcome (e.g., presence of traumatic brain injury). For each of these datasets, patients are divided into two natural groups: age <2 y and ≥ 2 y. This age-based threshold is commonly used for emergency-based diagnostic strategies (65), because it follows a natural stage of development, including a child’s ability to participate in their care (e.g., ability to verbally communicate with their doctor). At the same time, natural variability in early childhood development also creates opportunities to share information across this threshold. These datasets are nonstandard for machine learning; as such, we spend considerable time cleaning, curating, and preprocessing these features along with medical expertise included in the authorship team.^{¶¶} 60% of the data is used for training, 20% for tuning hyperparameters (including estimation of each patient’s group-membership $\mathbb{P}(\text{age} < 2 \text{ y} | X)$), and 20% for evaluating test performance.

4.3. Prediction Metrics. Prediction performance is measured by comparing the specificity of a model when sensitivity is constrained to be above a given threshold, chosen to be {92%, 94%, 96%, 98%}. High levels of sensitivity are crucial for CDIs so as to avoid potentially life-threatening false negatives (i.e., missing a diagnosis). For a given sensitivity threshold, maximizing specificity helps prevent false positives, which in turn reduces resource utilization and spares patients from the harmful effects of medical procedures such as radiation exposure from a computed tomography scan.

4.4. Baseline Methods. We compare FIGS and G-FIGS to two baselines: CART (8) and Tree-Alternating Optimization (TAO) (67). For each baseline, we either i) fit one model to all the training data or ii) fit a separate model to each group (denoted with -SEP), i.e., one to the patients with age <2 y and one for the patients with age ≥ 2 y. Additionally, for CART, we also fit a model in the style of G-FIGS, denoted as G-CART. Limits on the total number of splits for each model are varied from 2 to 16 splits, yielding a range of interpretable models.^{###} Details of hyperparameter selection are in [SI Appendix, section S6](#).

^{§§}This is methodologically analogous to a propensity score in the causal inference literature.
^{¶¶}Details, along with the openly released clean data can be found in [SI Appendix, section S6](#).
^{###}The choice of 16 splits is somewhat arbitrary, but we find that among 643 popular CDIs on [mdcalc](#), 93% contain no more than 16 splits and 95% contain no more than 20 splits.

Table 2. Clinical decision datasets for traumatic brain injury (TBI) (65), intra-abdominal injury (IAI) (64), and cervical spine injury (CSI) (66)

Name	Patients	Features	Outcome (count)	Outcome (%)
TBI	42,428	61	376	0.9
IAI	12,044	21	203	1.7
CSI	3,313	34	540	16.3

4.5. FIGS and G-FIGS Predict Well. Table 3 shows the prediction performance of FIGS, G-FIGS, and baseline methods. Further, we report the prediction performance of all methods for each age group separately (i.e., age <2 y and age ≥ 2 y) in [SI Appendix, Table S2 and S3](#) respectively. For high levels of sensitivity, G-FIGS generally improves the model’s specificity against the baselines. Further, G-FIGS also improves specificity for each age group, often outperforming both the models that fit all the data as well as the model that fits data for each group separately. This suggests that using group-specific models that incorporates information across the entire dataset can lead to better prediction in a heterogeneous population.

4.6. Features Used by G-FIGS Match Medical Knowledge. Fig. 3 shows the G-FIGS model on the cervical spinal injury (CSI) dataset for patients with age <2 y, while [SI Appendix, section S6](#) shows G-FIGS models for patients with age ≥ 2 y, and the other two datasets. The features used by the learned model match medical domain knowledge and mostly agree with previous work, e.g., features such as *focal neurologic signs*, *altered mental status*, and *torticollis* are all known to increase the risk of CSI (66). Further, features unique to each group largely relate to the age cutoff; the <2 y age group features include those that clinicians can assess without asking the patient (e.g., *substantial torso injury*), while that of the ≥ 2 y age group features (visualized in [SI Appendix, Fig S10](#)) require verbal responses (*neck pain*, *head pain*). This reflects medical intuition that nonverbal features are often more reliable features in the <2 y age group. Note that G-FIGS predicts presence of substantial torso injury or axial load to the head decreases the chance of CSI. We believe that this is a result of strong covariate imbalance, and corresponding subgroup sizes. For example, there are only five patients who have the combination of variables in tree 1 as well as substantial torso injury. Small subgroup sizes result in noisy residuals resulting in FIGS producing unintuitive splits. Investigating the source of these splits further, and methods to overcome the instability of noisy residuals remains as important future work.

4.7. G-FIGS Disentangles Clinical Risk Factors. Fig. 3 visualizes the G-FIGS and G-CART model for the <2 y age group on the CSI dataset. Both G-FIGS and G-CART utilize almost all the same features (apart from *Axial load to head*). However, unlike G-CART, G-FIGS disentangles risk factors with each tree representing a clinical domain: The top tree identifies signs and symptoms, the middle tree corresponds to the mode of injury and how the patient arrived at the hospital, and the bottom tree assesses the overall severity of the patient’s injury patterns and any associated injuries that may be present. By disentangling clinical domains, the fitted G-FIGS model not only has stronger prediction performance than a single-tree model (Table 2) but also provides a more medically intuitive framework for clinicians to use when assessing and treating patients. We provide another

Table 3. Best test-set specificity when sensitivity is constrained to be above a given level

Sensitivity level:	Traumatic brain injury				Cervical spine injury				Intra-abdominal injury			
	92%	94%	96%	98%	92%	94%	96%	98%	92%	94%	96%	98%
TAO	6.2	6.2	0.4	0.4	41.5	21.2	0.2	0.2	0.2	0.2	0.0	0.0
TAO-SEP	26.7	13.9	10.4	2.4	32.5	7.0	5.4	2.5	12.1	8.5	2.0	0.0
CART	20.9	14.8	7.8	2.1	38.6	13.7	1.5	1.1	11.8	2.7	1.6	1.4
CART-SEP	26.6	13.8	10.3	2.4	32.1	7.8	5.4	2.5	11.0	9.3	2.8	0.0
G-CART	15.5	13.5	6.4	3.0	38.5	15.2	4.9	3.9	11.7	10.1	3.8	0.7
FIGS	23.8	18.2	12.1	0.4	39.1	33.8	24.2	16.7	32.1	13.7	1.4	0.0
FIGS-SEP	39.9	19.7	17.5	2.6	38.7	33.1	20.1	3.9	18.8	9.2	2.6	0.9
G-FIGS	42.0	23.0	14.7	6.4	42.2	36.2	28.4	15.7	29.7	18.8	11.7	3.0

G-FIGS provides the best performance overall in the high-sensitivity regime. -SEP models fit a separate model to each group, and generally outperform fitting a model to the entire dataset. G-CART follows the same approach as G-FIGS but uses weighted CART instead of FIGS for each final group model. Averaged over 10 random data splits into training, validation, and test sets, with hyperparameters chosen independently for each split. Bolding indicates best performing algorithm.

example of the ability of FIGS to disentangle risk factors on a diabetes classification dataset in [SI Appendix, section S7](#). While these results highlight the promise of our approach, we note that there is no a priori reason that the true underlying data-generating process can be neatly disentangled by a tree-sum model. Instead, trust in the veracity of such a model should be based on good prediction performance as well as coherence with domain knowledge. In more detail, practitioners should follow the predictive, descriptive, and relevance framework for investigating real-world scientific problems using interpretable machine learning models (6).

4.8. Stability Analysis of Learned CDI. As discussed earlier, the PCS framework argues that stability to “reasonable” data perturbations is a key prerequisite for interpretability and deployment of machine learning methods in high-stakes domains. Here, we investigate the stability of G-FIGS on the CSI dataset. Specifically, we introduce noise by randomly swapping a percentage $p \in \{1\%, 2.5\%, 5\%\}$ of labels y . For each value of P , stability of G-FIGS is measured by comparing the similarity of the features selected in the model trained on the perturbed data to the model displayed in Fig. 3. Similarity of features is measured via the Jaccard distance. Further details of our experiments, and our results can be found in [SI Appendix, section S6](#). Our results show that G-FIGS is stable, and learns a similar model for each age group for all values of P .

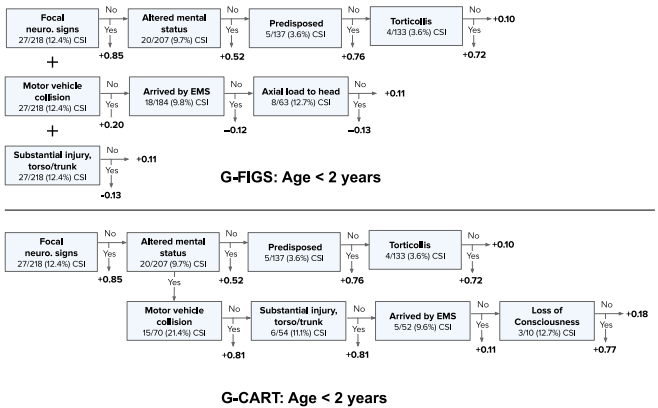


Fig. 3. CDI learned by G-FIGS and G-CART on the CSI dataset for patients younger than 2 y old. Both G-FIGS and G-CART use features that match medical knowledge. However, unlike G-CART, G-FIGS disentangles risk factors, with each tree representing a distinct clinical domain.

5. Theoretical Investigations

Theoretical investigations are performed to better understand the properties of FIGS and tree-sum models. Specifically, we show (under some oracle conditions) that if the regression function has an additive decomposition, then tree-sum models and FIGS are able to achieve optimal generalization upper bounds and disentangle additive components. This section summarizes these theoretical results, with formal statements and proof details to [SI Appendix, section S7](#).

5.1. Oracle Generalization Upper Bounds. Previous work has established *all* single-tree models (i.e., no matter their construction) have a squared error generalization error lower bound of $\Omega(n^{-2/(d+2)})$ when fitted to smooth additive models (13). To demonstrate the utility of tree-sum models in capturing additive structure, we provide generalization upper bounds of tree-sum models when their structure is chosen by an oracle. Specifically, we consider the typical supervised learning set-up: $y = f(\mathbf{x}) + \epsilon$, where $\mathbf{x}$ is a random variable on $[0, 1]^d$ and $\mathbb{E}[\epsilon | \mathbf{x}] = 0$. Assume $f(\mathbf{x}) = \sum_{k=1}^K f_k(\mathbf{x}_{I_k})$, where $I_1 \dots I_K$ are disjoint blocks of features, and $\mathbf{x}_{I_k}$ denotes the subvector of $\mathbf{x}$ comprising coordinates in I_k . Under this set-up, if each component function f_k is smooth, and blocks of features are independent (i.e., $\mathbf{x}_{I_j} \perp \mathbf{x}_{I_k}$ for $j \neq k$), then there exists a tree-sum model such that its squared error generalization upper bound scales as $O(Kn^{-2/(d_{\max}+2)})$, where $d_{\max} = \max_k |I_k|$. It is instructive to consider two extreme cases: If $|I_k| = 1$ for each k , the upper bound scales as $O(dn^{-2/3})$. On the other hand if $K = 1$, we have an upper bound of $O(n^{-2/(d+2)})$. Both bounds match the well-known minimax rates for their respective inference problems (68). See Theorem 2 for the formal statement.

5.2. FIGS Performs Disentanglement. One potential reason for the success of FIGS is its ability to disentangle additive structure. Under the generative model discussed above, if FIGS splits nodes using population quantities (i.e., in the large-sample limit), then the set of features split upon in each fitted tree $\hat{f}_k$ is contained within I_k . The precise theorem statement can be found in Theorem 1 in [SI Appendix, section S7](#). By disentangling additive components, FIGS is able to avoid duplicate subtrees, leading to a more parsimonious model with better performance since number of splits is proportional to variance of the model (13). We provide (partial) empirical justification of disentanglement in real-world datasets by showing that FIGS helps to reduce the number of possibly redundant and repeated splits often observed

in CART models (SI Appendix, section A). As discussed earlier, we emphasize that disentanglement does not necessarily reflect the underlying data-generating process. See Section 4 for a larger discussion regarding interpreting disentanglement.

6. Bagging-FIGS

We now introduce *Bagging-FIGS*, which averages the predictions from an ensemble of FIGS models, and makes use of the same variance reduction strategies as random forests. Specifically, i) Each FIGS model fit on a bootstrap resampled dataset, and ii) at every iteration of each FIGS model in the ensemble, the algorithm chooses the next split only from a random subset of the original features. The effect of both these strategies has been studied empirically and theoretically (69–72). We note that Bagging-FIGS is not inherently interpretable, and so we only focus on its prediction performance, where it performs comparable to other tree-based ensembles such as random forests and XGBoost.

6.1. Baseline Methods and Settings. We compare the prediction performance of FIGS and Bagging-FIGS to that of four other algorithms: random forest, XGBoost, and penalized iteratively reweighted least squares (PIRLS) on the log-likelihood of a generative additive model. All algorithms are fit using default settings [We use the implementation of PIRLS in pygam (73), with 20 splines term for each feature]. The number of features subsetted is a tuning parameter for random forest and Bagging-FIGS. For both algorithms, we set this to be $d/3$ for regression datasets and $\sqrt{d}$ for classification datasets. These are the default choices for RF. We fit Bagging-FIGS with 100 base FIGS estimators.

6.2. Bagging-FIGS Performs Comparably to Random Forest and XGBoost. Fig. 4 shows the generalization performance of all the methods on the various real-world datasets introduced in Sections 3 and 4. Bagging-FIGS improves upon the performance of FIGS via bagging and feature subsetting. In addition, Bagging-FIGS achieves comparable AUC to both XGBoost and random forest (on average improving over XGBoost by 0.015 and random forest by 0.013). For some datasets (e.g. *IAI pecarn*, *Recidivism*), Bagging-FIGS outperforms both baselines.

7. Discussion

FIGS is a natural extension to CART which achieves improved predictive performance over popular baseline tree-based methods across a wide array of datasets while maintaining interpretability. Furthermore, when the number of splits is unconstrained, an ensemble version of FIGS has prediction performance that compares favorably to random forest and XGBoost.

As a case study, we have shown how FIGS and G-FIGS can make an important step toward interpretable modeling of heterogeneous data in the context of high-stakes clinical decision-making. The fitted CDI models here show promise, but require external clinical validation before potential use. While our current work only explores age-based grouping, the behavior of G-FIGS with temporal, geographical, or demographic splits could be studied as well. Additionally, there are many methodological extensions to explore, such as data-driven identification of input data groups and schemes for feature weighting in addition to instance weighting. FIGS has many other natural extensions, some of which we detail below.

7.1. Optimization. Instead of using a greedy approach, one may perform global optimization algorithm over the class of tree-sum models as has been done via dynamic programming techniques for single decision tree models (25, 74). FIGS could include other moves such as pruning in addition to just adding more splits. This could also be embedded in a Bayesian framework, similar to Bayesian Additive Regression Trees (43).

7.2. Regularization. Using cross-validation (CV) to select the number of splits tends to select larger models. Future work can explore criteria related to Bayesian Information Criteria (75) or stability in combination with CV (76) for selecting this threshold based on data. In future work, one could also vary the total number of splits and number of trees separately, helping to build prior knowledge into the fitting process. FIGS could be penalized via novel regularization techniques, such as regularizing individual leaves or regularizing a linear model formed from the splits extracted by FIGS (23).

7.3. Learning Interactions. Interactions are known to be prevalent in biology and other fields, and therefore of key scientific interest (77). Random forests have been successfully applied to

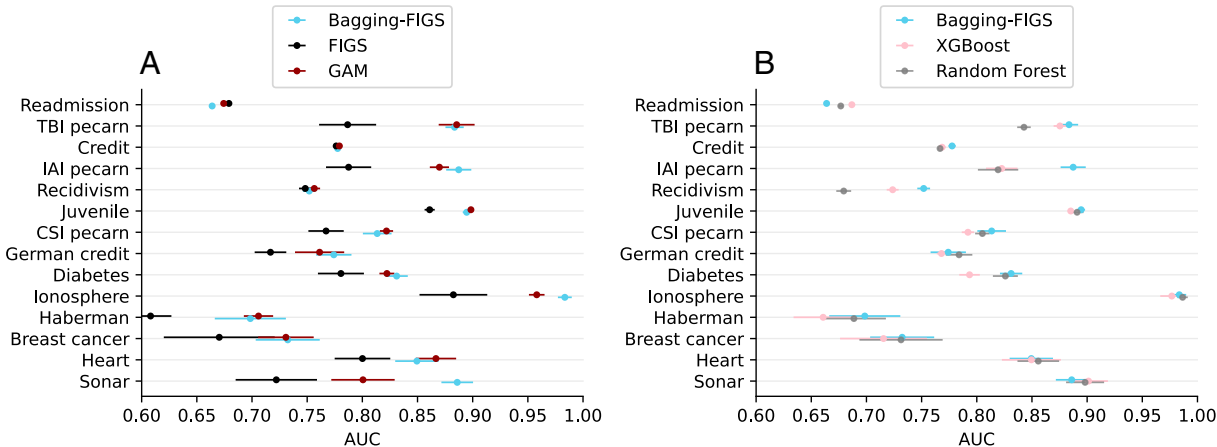


Fig. 4. Generalization performance across various real-world datasets with no constraints on the number of splits. (A) The performance of FIGS can usually be improved by fitting an ensemble of FIGS models using bagging and feature subsetting (Bagging-FIGS). The performance of Bagging-FIGS is comparable to that of a GAM. (B) Bagging-FIGS achieves comparable performance to both XGBoost and random forest. Error bars show SE of the mean, computed over 5 random data splits.

learn biologically meaningful interactions (14, 78, 79). However, since single decision trees are unable to disentangle additive structure, using random forests for interaction discovery might lead to a high false discovery rate. As a result, using FIGS in lieu of single decision trees might lead to improved interaction discovery. We leave this investigation to future work.

7.4. Model Class. The class of FIGS models could be further extended to include linear terms or allow for summations of trees to be present at split nodes, rather than just at the root.

We hope FIGS and G-FIGS can pave the way toward more transparent and interpretable modeling that can improve machine-learning practice, particularly in high-stakes domains such as medicine, law, and policy making.

Data, Materials, and Software Availability. Tabular data (i.e., .csv) containing benchmark supervised learning datasets and data for clinical decision-making have been deposited in two repositories: imodels-data at <https://doi.org/10.5281/zenodo.8139590> and OpenML at <https://doi.org/10.1145/2641190.2641198>.

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